The Collected Mathematical Papers

of

Arthur Cayley, Sc.D., F.R.S.

Sadlerian Professor of Pure Mathematics in the University of Cambridge.

Vol. VII.

Cambridge: at the University Press

1894

Advertisement

The present volume contains 69 papers numbered 417 to 485, published for the most part in the years 1866 to 1872; they include a series of astronomical papers published in the Memoirs and Monthly Notices of the Royal Astronomical Society.

The Portrait in Volume VI. is from the Painting in Oil by Mr Lowes Dickinson in the year 1874, presented by the Subscribers to Trinity College, Cambridge, and now in the Hall of the College: the portrait in the present volume is a photograph of a pencil sketch by Mr Lowes Dickinson in the year 1893.

The Table for the seven volumes is

Vol. I.	Numbers 1 to 100.
Vol. II.	Numbers 101 to 158.
Vol. III.	Numbers 159 to 222.
Vol. IV.	Numbers 223 to 299.
Vol. V.	Numbers 300 to 383.
Vol. VI.	Numbers 384 to 416.
Vol. VII.	Numbers 417 to 485.

Contents

[417]

On the Locus of the Foci of the Conics which pass through Four Given Points.

[From the Philosophical Magazine, vol. XXXII. (1866), pp. 362–365.]

THE curve which is the locus of the foci of the conics which pass through four given points is, as appears from a general theorem of M. Chasles, a sextic curve having a double point at each of the circular points at infinity; and Professor Sylvester, in his "Supplemental Note on the Analogues in Space to the Cartesian Ovals *in plano*" (Phil. Mag., May 1866), has further remarked that the lines (eight in all) joining the circular points at infinity with any one of the four points are all of them double tangents of the curve; whence each of these points is a focus (more accurately a quadruple focus) of the curve. It is to be added that, besides the circular points at infinity, the curve has 6 double points (3 of these are the centres of the quadrangles formed by the 4 points), in all 8 double points; the class is therefore = 14. Hence also the number of tangents to the curve from a circular point at infinity is = 10; viz. these are the 4 double tangents each reckoned twice, and 2 single tangents; and the theoretical number of foci is = 100; viz. we have

16 quadruple foci, or intersections of a double tangent by a double tangent	$16 \times 4 = 64$
16 double foci, or intersections of a double tangent by a single tangent	$16 \times 2 = 32$
4 single foci, or intersections of a single tangent by a single tangent	4
	100

To verify the foregoing results, consider any two given points I, J, and the series of conics which pass through four given points A, B, C, D; we have thus a curve the locus of the intersections of the tangents from I and the tangents from J to any conic of the series; which curve, if I, J are the circular points at infinity, is the required curve of foci. Taking $U + \lambda V = 0$ for the equation of a conic of the series, the pair of tangents from I is given by an equation of the form

$$(\lambda,\ 1)^2(x, y, z)^2 = 0,$$

and the pair of tangents from J by an equation of the like form

$$(\lambda,\ 1)^2(x, y, z)^2 = 0;$$

and by eliminating λ from these equations, we obtain the equation of the required curve. This in the first instance presents itself as an equation of the eighth order; but it is to be observed that in the series of conics there are two conics each of them touching the line IJ, and that, considering the tangents drawn to either of these conics, the line IJ presents itself as part of the locus; that is, the line IJ twice repeated is part of the locus; and the residual curve is thus of the order $8 - 2, = 6$; that is, the required curve is of the order 6. The consideration of the same two conics shows that each of the points I, J is a double point on the curve. Moreover, by taking for the conic any one of the line-pairs through the four points, it appears that each of the points $(AB.CD)$, $(AC.BD)$, $(AD.BC)$ is a double point on the curve: this establishes the existence of five double points. The two conics of the series which touch the line IA are a single conic taken twice, and the consideration of this conic shows that the line IA is a double tangent to the curve; similarly each of the eight lines $I(A, B, C, D)$ and $J(A, B, C, D)$ is a double tangent to the curve.

Instead of seeking to establish directly the existence of the remaining three double points, the easier course is to show that, besides the four double tangents from I, the number of tangents from I to the curve is $= 2$; for, this being so, the total number of tangents from I to the curve will be $(2 \times 4 + 2 =)\, 10$; that is, I being a double point, the class of the curve is $= 14$; and assuming that the depression $(6 \times 5 - 14 =)\, 16$ in the class of the curve is caused by double points, the number of double points will be $= 8$. But observing that in the series of conics there is one conic which passes through J, so that the tangents from J to this conic are the tangent at J twice repeated, then it is easy to see that the tangents from I to this conic, at the points where they meet the tangent at J, touch the required curve, and that these two tangents are in fact (besides the double tangents) the only tangents from I to the curve; that is, the number of tangents from I to the curve is $= 2$.

Considering I, J as the circular points at infinity, and writing A, B, C, D to denote the squared distances of a point P from the four points A, B, C, D respectively, then, as remarked by Professor Sylvester, the equation

$$\lambda\sqrt{A} + \mu\sqrt{B} + \nu\sqrt{C} + \pi\sqrt{D} = 0$$

(where λ, μ, ν, π are constants) is in general a curve of the order 8; but the ratios $\lambda : \mu : \nu : \pi$ may be so determined that the order of the curve in question shall be $= 6$; the resulting curve of the order 6 is (not one of a group of curves, but the very curve) the locus of the foci of the conics through the four points. And the determination of the ratios $\lambda : \mu : \nu : \pi$ is in fact quite simple; for writing

$$\begin{aligned} A &= (x-a)^2 + (y-\alpha)^2 \\ &= \rho^2 - 2(ax + \alpha y) + \&\text{c.}, \quad (\text{if } \rho^2 = x^2 + y^2), \end{aligned}$$

and therefore

$$\sqrt{A} = \rho\sqrt{1 - \cdots},$$

with similar values for $\sqrt{B}$, $\sqrt{C}$, $\sqrt{D}$, it is easy to see that, considering λ, μ, ν, π as standing for $\pm\sqrt{A}$, $\pm\sqrt{B}$, $\pm\sqrt{C}$, $\pm\sqrt{D}$ respectively, the conditions for the reduction to the order 6 are

$$\begin{aligned} \lambda + \mu + \nu + \pi &= 0, \\ \lambda a + \mu b + \nu c + \pi d &= 0, \\ \lambda\alpha_1 + \mu b_1 + \nu c_1 + \pi d_1 &= 0, \end{aligned}$$

and hence that the required equation of the curve of foci is

$$\Sigma \begin{vmatrix} 1, & 1, & 1 \\ b, & c, & d \\ b_1, & c_1, & d_1 \end{vmatrix} \sqrt{(x-a)^2 + (y-\alpha_1)^2} = 0,$$

or, as this may also be written,

$$\Sigma \pm (B,\, C,\, D)\sqrt{A} = 0,$$

where $(B,\, C,\, D)$, &c. are the areas of the triangles B, C, D, &c.

I remark, in conclusion, that the number of conditions to be satisfied in order that a curve may have for double points two given points I, J, may have besides six double points, and may have for double tangents eight given lines, is $(3 + 3 + 6 + 16 =)\, 28$; the number of constants contained in the general equation of the order 6 is $= 27$. The conditions that a curve of the order 6 shall have for double points two given points I, J, shall besides have six double points, and shall have for double tangents four given lines through I and four given lines through J, are more than sufficient for the determination of the sextic curve; and the existence of a sextic curve satisfying these conditions is therefore a theorem.

In the case where the points I, J lie on a conic of the series, the consideration of this conic shows that the curve has a ninth double point, the pole of the line IJ in regard to the conic in question: in this case the sextic curve, as is known, breaks up into two cubic curves. [It need not do so, for a proper sextic curve may have nine (or indeed ten) double points.]

P.S. In general the curve $\lambda\sqrt{A} + \mu\sqrt{B} + \nu\sqrt{C} + \pi\sqrt{D} = 0$ has (exclusively of multiple points at infinity) six double points; viz. these are situate at the intersections of the pairs of circles,

$$(\lambda\sqrt{A} + \mu\sqrt{B} = 0,\ \ \nu\sqrt{C} + \pi\sqrt{D} = 0),$$
$$(\lambda\sqrt{A} + \nu\sqrt{C} = 0,\ \ \mu\sqrt{B} + \pi\sqrt{D} = 0),$$
$$(\lambda\sqrt{A} + \pi\sqrt{D} = 0,\ \ \mu\sqrt{B} + \nu\sqrt{C} = 0).$$

In the case of the curve of foci, the first, second, and third pairs of circles intersect respectively in the points $(AB.CD)$, $(AC.BD)$, $(AD.BC)$, which, as mentioned above, are double points on the curve; and they besides intersect in three other points, which are the other three double points mentioned above.

Professor Sylvester reminds me that he mentioned to me in conversation that he had himself obtained the foregoing equation $\Sigma \pm (B, C, D)\sqrt{A} = 0$, for the locus of the foci of the conics which pass through the four points A, B, C, D.

Cambridge, October 10, 1866.

[418]

A Remark on Differential Equations.

[From the Philosophical Magazine, vol. XXXII. (1866), pp. 379–381.]

CONSIDER a differential equation $f(x, y, p) = 0$, of the first order, but of the degree n, where f is a rational and integral function of (x, y, p) not rationally decomposable into factors: the integral equation contains an arbitrary constant c, and represents therefore a system of curves, for any one of which curves the differential equation is satisfied: the differential equation is assumed to be such that the curves are algebraical curves. The curves in question may be considered as undecomposable curves; in fact, if the curve $U^\alpha V^\beta W^\gamma \ldots = 0$ (composed of the undecomposable curves $U = 0$, $V = 0$, $W = 0$,...) satisfies the differential equation, then either the curves $U = 0$, $V = 0$, $W = 0$,... each satisfy the differential equation, and instead of the curve $U^\alpha V^\beta W^\gamma \ldots = 0$ we have thus the undecomposable curves $U = 0$, $V = 0$, $W = 0$,... each satisfying the differential equation; or if any of these curves, for instance $W = 0$, &c., do not satisfy the differential equation, then W^γ, &c. are mere extraneous factors which may and ought to be rejected, and instead of the original curve $U^\alpha V^\beta W^\gamma \ldots = 0$, we have the undecomposable curves $U = 0$, $V = 0$ satisfying the differential equation. Assuming, as above, the existence of an algebraical solution, this may always be expressed in the form $\phi(x, y, c) = 0$, where ϕ is a rational and integral function of (x, y, c), of the degree n as regards the arbitrary constant c: this appears by the consideration that for given values (x_0, y_0) of (x, y) the differential equation and the integral equation must each of them give the same number of values of p. It is to be observed that ϕ regarded as a function of (x, y, c) cannot be rationally decomposable into factors; for if the equation were $\phi = \Phi\Psi \ldots = 0$, Φ, Ψ, &c. being each of them rational and integral functions of (x, y, c), then the differential equation would be satisfied by at least one of the equations $\Phi = 0$, $\Psi = 0$, ... that is, by an equation of a degree less than n in the arbitrary constant c.

But the equation $\phi(x, y, c) = 0$ is not of necessity the equation of an undecomposable curve, and the undecomposable curve which constitutes the proper solution of the differential equation cannot always be represented by an equation of the form in question. For although ϕ regarded as a function of (x, y, c) is not rationally decomposable into factors, yet it may very well happen that ϕ regarded as a function of (x, y) is rationally decomposable into factors (geometrically the sections by the planes $z = c$ of the undecomposable surface $\phi(x, y, z) = 0$ may each of them be composed of two or more distinct curves); and assuming that the function ϕ is thus decomposed into its prime factors, then each factor equated to zero gives an undecomposable curve satisfying the differential equation, and constituting the proper solution thereof.

It may be observed that, by the foregoing process of decomposition, we sometimes reduce the original equation $\phi(x, y, c) = 0$ into a like equation $\phi_1(x, y, c_1) = 0$ of a more simple form. Thus, for instance, if we have $\phi(x, y, c) = U^2 - c = 0$, U being a rational and integral function of (x, y), then instead of $U^2 - c = 0$ we have the equations $U + \sqrt{c} = 0$, $U - \sqrt{c} = 0$, each of which is an equation of the form $U - c_1 = 0$, or we pass from the original equation $\phi(x, y, c) = U^2 - c = 0$ to the simplified equation

$$\phi_1(x, y, c_1) = U - c_1 = 0.$$

Again, to take a somewhat more complicated instance, if the given integral equation be

$$\phi(x, y, c) = U^4 + c^2V^4 + (c + 1)^2W^4 - 2c\,U^2V^2 - 2(c + 1)\,U^2W^2 - 2c(c + 1)\,V^2W^2 = 0,$$

then the equation $U^2 + V^2\sqrt{c} + W^2\sqrt{c + 1} = 0$, writing therein $\sqrt{c} = \dfrac{c_1^2 + 1}{c_1^2 - 1}$, and therefore $\sqrt{c + 1} = \dfrac{2c_1}{c_1^2 - 1}$, becomes

$$U(c_1^2 - 1) + V \cdot 2c_1 + W(c_1^2 + 1) = 0;$$

so that we pass from the original equation $\phi(x, y, c)$ to the simplified equation

$$\phi_1(x, y, c_1) = U(c_1^2 - 1) + V \cdot 2c_1 + W(c_1^2 + 1) = 0.$$

But observe that the possibility of the rationalization depends on the form of the radicals $\sqrt{c}$ and $\sqrt{c+1}$; if we had had $\sqrt{c}$ and $\sqrt{c^2+1}$ (or c and $\sqrt{c^4+1}$), the rationalization could not have been effected.

Returning to the case of an integral equation $\phi(x, y, c) = 0$, where ϕ regarded as a function of (x, y) is decomposable into factors, then equating to zero any one of the prime factors of ϕ, we obtain an integral equation $\psi(x, y, c_1, c_2, \ldots c_k) = 0$, where $c_1, c_2, \ldots c_k$ are irrational functions (not of necessity representable by radicals, and without any superior limit to the number of these functions) of c: here ψ regarded as a function of (x, y) is of course undecomposable, and the equation $\psi(x, y, c_1, c_2, \ldots c_k) = 0$ belongs to the undecomposable curve which is the proper solution of the differential equation. The result may be stated under a quasi-geometrical form; viz. regarding $c_1, c_2, \ldots c_k$ as the coordinates of a point in k-dimensional space, then as these are functions of the single parameter c, the point to which they belong is an arbitrary point on a certain curve or $(k-1)$fold locus C in the k-dimensional space. And this curve must be such that to given values of (x, y) there shall correspond n points on the curve; that is, treating (x, y) as constants, the surface or onefold locus $\psi(x, y, c_1, c_2, \ldots c_k) = 0$, and the curve or $(k-1)$fold locus C, shall meet in n points. The conclusion stated in the foregoing quasi-geometrical form is, that the solution of the differential equation may be exhibited in the form $\psi(x, y, c_1, c_2, \ldots c_k) = 0$; viz. ψ is a rational and integral function of $(x, y, c_1, c_2, \ldots c_k)$, where $(c_1, c_2, \ldots c_k)$ are the coordinates of an arbitrary or variable point on a curve or $(k-1)$fold locus C in a k-dimensional space, which curve meets the surface or onefold locus $\psi(x, y, c_1, c_2, \ldots c_k)$ in n points, and where ψ regarded as a function of (x, y) is not rationally decomposable into factors.

Cambridge, October 13, 1866.

[419]

A Theorem on Differential Operators.

[From a paper by Prof. Sylvester, "Note on the Test Operators which occur in the Calculus of Invariants, &c.," Philosophical Magazine, vol. XXXII. (1866), pp. 461–472, see p. 471.]

THE paper concludes with an Observation from Professor Cayley as follows:

"In the case of two variables, if

$$P_1 = (ax + by)\,\frac{d}{dx} + (cx + dy)\,\frac{d}{dy},$$

then in the notation of matrices,

$$P_1 = \begin{pmatrix} a, & b \\ c, & d \end{pmatrix} (x,\, y) \left(\frac{d}{dx},\, \frac{d}{dy} \right),$$

$$P_2 = \tfrac{1}{2} \begin{pmatrix} a, & b \\ c, & d \end{pmatrix}^2 (x,\, y) \left(\frac{d}{dx},\, \frac{d}{dy} \right),$$

$$P_3 = \tfrac{1}{6} \begin{pmatrix} a, & b \\ c, & d \end{pmatrix}^3 (x,\, y) \left(\frac{d}{dx},\, \frac{d}{dy} \right);$$

whence also

$$P \cdot P_1 = P_1 \cdot P + P_2 = \tfrac{1}{2} \begin{pmatrix} a, & b \\ c, & d \end{pmatrix}^2 (x,\, y) \left(\frac{d}{dx},\, \frac{d}{dy} \right) = 3P_2,$$

which accords with your theorem,

$$E_1 * E_2 * = E_2 * E_1 * = E_3 E_0 * + 3E_2 * ."$$

I have taken the liberty of writing in the above $\frac{d}{dx}, \frac{d}{dy}$ for δ_x, δ_y, and P for δ in the original. It will be useful to bear in mind that in any operator such as E_1* or E_2*, the asterisk forms an integral part of the symbol. Thus $E_1 * E_2*$, if we choose, may be written under the form of E_1* multiplied by E_2*, i.e. $(E_1*) \times (E_2*)$, where the cross is the sign of ordinary algebraical multiplication.

[420]

On Riccati's Equation.

[From the Philosophical Magazine, vol. XXXVI. *(1868), pp. 348–351.]*

THE following is, it appears to me, the proper form in which to present the solution of Riccati's equation.

The equation may be written

$$\frac{dy}{dx} + y^2 = x^{2q-2},$$

which is integrable by algebraic and exponential functions if $(2i+1)q = \pm 1$, i being zero, or a positive integer. To effect the integration, writing $y = \frac{1}{u}\frac{du}{dx}$, we have

$$\frac{d^2u}{dx^2} = x^{2q-2}\,u.$$

The peculiar advantage of this well-known transformation has not (so far as I am aware) been explicitly stated; it puts in evidence the form under which the sought-for function y contains the constant of integration. In fact if $u = P$, $u = Q$ be two particular solutions of the equation in u, then the general solution is $u = CP + DQ$; and denoting by P', Q' the derived functions, the value of y is

$$y = \frac{CP' + DQ'}{CP + DQ},$$

showing the form under which the constant of integration $C \div D$ is contained in y.

To complete the solution, assume

$$u = ze^{\frac{1}{q}x^q};$$

we find

$$\frac{d^2z}{dx^2} + 2x^{q-1}\frac{dz}{dx} + (q-1)\,x^{q-2}\,z = 0;$$

considering first the particular integral of the form

$$z = A + Bx^q + Cx^{2q} + Dx^{3q} + \&\text{c.},$$

we find that the equation will be satisfied if

$$\begin{aligned}
(\ q-1)\,A + \ \ q\,(\ q-1)\,B &= 0,\\
(3q-1)\,B + 2q\,(2q-1)\,C &= 0,\\
(5q-1)\,C + 3q\,(3q-1)\,D &= 0,\\
(7q-1)\,D + 4q\,(4q-1)\,E &= 0,\\
&\&\text{c.}
\end{aligned}$$

If $A = 1$, this is

$$\begin{aligned}
A &= \ \ 1,\\
B &= -\frac{q-1}{q(q-1)},\\
C &= +\frac{(q-1)(3q-1)}{q(q-1)\,2q(2q-1)},\\
D &= -\frac{(q-1)(3q-1)(5q-1)}{q(q-1)\,2q(2q-1)\,3q(3q-1)},\\
&\&\text{c.},
\end{aligned}$$

where it is to be noticed that the series may be considered to stop so soon as there is in the numerator a factor $= 0$. For instance, if $5q - 1 = 0$, then if the particular integral had been assumed to be $z = A + Bx^q + Cx^{2q}$, the only conditions to be satisfied by the coefficients are the first and second equations giving the foregoing values of A, B, C. It is immaterial that the analytical expressions of F and the subsequent coefficients contain in the denominators the evanescent factor $5q - 1$; the coefficients after C do not ever come into consideration.

Thus if $(2i+1)q = +1$, the series terminates, and we have for u the finite particular solution

$$u = P = \left(1 - \frac{q-1}{q(q-1)}\,x^q + \frac{(q-1)(3q-1)}{q(q-1)\,2q(2q-1)}\,x^{2q} - \&\text{c.}\right) e^{\frac{1}{q}x^q},$$

and it is easy to see that we may herein change the sign of x^q, thereby obtaining another finite particular solution,

$$u = Q = \left(1 + \frac{q-1}{q(q-1)}\,x^q + \frac{(q-1)(3q-1)}{q(q-1)\,2q(2q-1)}\,x^{2q} + \&\text{c.}\right) e^{-\frac{1}{q}x^q}.$$

Reverting to the equation in z, we have next a particular solution of the form

$$z = Ax + Bx^{q+1} + Cx^{2q+1} + Dx^{3q+1} + \&\text{c.},$$

giving between the coefficients the relation

$$\begin{aligned}
(\ \ q+1)\,A + (\ \ q+1)\,q\,B &= 0,\\
(3q+1)\,B + (2q+1)\,2q\,C &= 0,\\
(5q+1)\,C + (3q+1)\,3q\,D &= 0,\\
(7q+1)\,D + (4q+1)\,4q\,E &= 0,\\
&\&\text{c.}
\end{aligned}$$

If $A = 1$, we have

$$\begin{aligned}
A &= \ \ 1,\\
B &= -\frac{q+1}{(q+1)\,q},\\
C &= +\frac{(q+1)(3q+1)}{(q+1)\,q\,(2q+1)\,2q},\\
D &= -\frac{(q+1)(3q+1)(5q+1)}{(q+1)\,q\,(2q+1)\,2q\,(3q+1)\,3q},\\
&\&\text{c.},
\end{aligned}$$

where, as in the former case, the series is considered to terminate as soon as there is an evanescent factor in the numerator, without any regard to the subsequent coefficients which contain in the denominators the same evanescent factor. [In particular, $q = -1$, we have the solution $z = x$.]

Hence if we have $(2i+1)q = -1$, the series terminates, and we have for u the finite particular solution,

$$u = P = x\left(1 - \frac{q+1}{(q+1)\,q}\,x^q + \frac{(q+1)(3q+1)}{(q+1)\,q\,(2q+1)\,2q}\,x^{2q} - \&\text{c.}\right) e^{\frac{1}{q}x^q},$$

from which, changing the sign of x^q, we deduce the other finite particular solution,

$$u = Q = x\left(1 + \frac{q+1}{(q+1)\,q}\,x^q + \frac{(q+1)(3q+1)}{(q+1)\,q\,(2q+1)\,2q}\,x^{2q} + \&\text{c.}\right) e^{-\frac{1}{q}x^q}.$$

Hence, in the equation

$$\frac{dy}{dx} + y^2 = x^{2q-2},$$

where $q(2i+1) = \pm 1$, we have (writing $D = 1$)

$$y = \frac{CP' + Q'}{CP + Q},$$

where C is the constant of integration, P, Q are finite series as above, and P', Q' are the derived functions of P and Q. Writing successively $i = 0$, $i = 1$, $i = 2$, &c., we may tabulate the solutions:

$$\begin{array}{llll}
q + 2i = 1, & \frac{dy}{dx} + y^2 = 1, & P = e^{x}, & Q = e^{-x}, \\
& \frac{dy}{dx} + y^2 = x^{-4}, & P = xe^{-\frac{1}{x}}, & Q = xe^{\frac{1}{x}}, \\
& \frac{dy}{dx} + y^2 = x^{-\frac{8}{3}}, & P = (1 - 3x^{\frac{1}{3}})\, e^{3x^{\frac{1}{3}}}, & Q = (1 + 3x^{\frac{1}{3}})\, e^{-3x^{\frac{1}{3}}}, \\
& \frac{dy}{dx} + y^2 = x^{-\frac{4}{3}}, & P = (1 - 5x^{\frac{1}{5}} + \frac{25}{3}x^{\frac{2}{5}})\, e^{5x^{\frac{1}{5}}}, & Q = (1 + 5x^{\frac{1}{5}} + \frac{25}{3}x^{\frac{2}{5}})\, e^{-5x^{\frac{1}{5}}}, \\
& & & \text{\&c.}
\end{array}$$

It is hardly necessary to make the final step of calculating P' and Q' and substituting in y; but, as an example, take the above equation $\dfrac{dy}{dx} + y^2 = x^{-\frac{8}{3}}$: we have

$$y = \frac{-3x^{-\frac{1}{3}}\,(C\,e^{3x^{\frac{1}{3}}} + e^{-3x^{\frac{1}{3}}})}{C\,(1 - 3x^{\frac{1}{3}})\,e^{3x^{\frac{1}{3}}} + (1 + 3x^{\frac{1}{3}})\,e^{-3x^{\frac{1}{3}}}},$$

which is readily identified with the solution, p. 98 of Boole's *Differential Equations* (Cambridge, 1859).

Cambridge, September 29, 1868.

[421]

Note on the Solvibility of Equations by means of Radicals.

[From the Philosophical Magazine, vol. XXXVI. (1868), pp. 386, 387.]

In regard to the theorem that the general quintic equation of the nth order is not solvible by radicals, I believe that the proofs which have been given depend, or at any rate that a proof may be given that shall depend, on the following two lemmas:

I. A one-valued (or symmetrical) function of n letters is a perfect kth power, only when the kth root is a one-valued function of the n letters.

There is an exception in the case $k=2$, whatever be the value of n: viz. the product of the squares of the differences is a one-valued function, a perfect square; but its square root, or the product of the simple differences, is a two-valued function. It is in virtue of this exception that a quadric equation is solvible by radicals; we have the one-valued function $(x_1-x_2)^2$, the square of a two-valued function x_1-x_2, and thence the two roots are each expressible in the form

$$\tfrac{1}{2}\{x_1+x_2 \pm \sqrt{(x_1-x_2)^2}\}.$$

II. A two-valued function of n letters is a perfect kth power, only when the kth root is a two-valued function of the n letters.

There is an exception in the case $k=3$, when $n=3$ or 4: viz. for $n=3$ we have $(x_1+\omega x_2+\omega^2 x_3)^3$ (ω an imaginary cube root of unity) a two-valued function, and a perfect cube; whereas its cube root is the six-valued function $x_1+\omega x_2+\omega^2 x_3$. And similarly for $n=4$ we have, for instance,

$$\{x_1x_2+x_3x_4+\omega(x_1x_3+x_2x_4)+\omega^2(x_1x_4+x_2x_3)\}^3$$

a two-valued function, and a perfect cube, whereas its cube root is a six-valued function. And it is in virtue of this exception that a cubic or a quartic equation is solvible by radicals. But I assume that for $n>4$ the lemma is true without exception.

The course of demonstration would be something as follows: Imagine, if possible, the root of an equation expressed, by means of radicals, in terms of the coefficients; the expression cannot contain any radical such as $\sqrt[p]{X}$, $p>2$, where X is a one-valued (or rational) function of the coefficients, not a perfect pth power, for the reason that, expressing the coefficients in terms of the roots, such function $\sqrt[p]{X}$ is not a rational function of the roots; if it were so, by lemma I. it would be a one-valued (that is, a symmetrical) function of the roots; consequently a rational function of the coefficients, or X expressed in terms of the coefficients, would be a perfect pth power.

The expression may however contain a radical $\sqrt{X}$, X a one-valued (or rational) function of the coefficients, not a perfect square: viz. X may be any square function multiplied into that function of the coefficients which is equal to the product of the squared differences of the roots, or, say, multiplied into the discriminant; that is, we may have $X=Q^2\nabla$, or $\sqrt{X}=Q\sqrt{\nabla}$.

We have next to consider whether the expression can contain any radical $\sqrt[p]{X}$, where X, not being a rational function of the coefficients, is a function expressible by radicals. But the foregoing reasoning shows that if this be so, X cannot contain any radical other than the radical $\sqrt{Q^2\nabla}$ or $Q\sqrt{\nabla}$, as above; that is, X must be $P+Q\sqrt{\nabla}$, where P and Q are rational functions of the coefficients, and where we may assume that $P+Q\sqrt{\nabla}$ is not a perfect pth power of a function of the like form $P'+Q'\sqrt{\nabla}$. But then, expressing the coefficients in terms of the roots, we have $P+Q\sqrt{\nabla}$, a (rational) two-valued function of the roots; and there is no radical $\sqrt[p]{P+Q\sqrt{\nabla}}$, which is a rational function of the roots; for by lemma II., if such radical existed we should have $\sqrt[p]{P+Q\sqrt{\nabla}}$ a (rational) two-valued function of the roots; that is, it would be

$= P' + Q'\sqrt{\nabla}$, P' and Q' one-valued (symmetrical) functions of the roots, consequently rational functions of the coefficients; or $P + Q\sqrt{\nabla}$ would be a perfect pth power $(P' + Q'\sqrt{\nabla})^p$.

The conclusion is that for $n > 4$ there is not (besides the function $P + Q\sqrt{\nabla}$) any function of the coefficients, expressible by means of radicals, which, when the coefficients are expressed in terms of the roots, will be a rational function of the roots, and consequently there is no possibility of expressing the roots in terms of the coefficients by means of radicals.

Cambridge, October 1, 1868.

[422]

On the Geodesic Lines on an Oblate Spheroid.

[From the Philosophical Magazine, vol. XL. *(1870), pp. 329–340.]*

THE theory of the geodesic lines on an oblate spheroid of any excentricity whatever was investigated by Legendre[1]; and the general course of them is well known, viz. each geodesic line undulates between two parallels equidistant from the equator (being thus either a closed curve, or a curve of indefinite length, according to the distance between the two parallels): at a point of contact with the parallel the curve is, of course, at right angles to the meridian; say this is V, a vertex of the geodesic line, and let the meridian through V meet the equator in A; the geodesic line proceeds from V to meet the equator in a point N, the node, where AN is at most $= 90°$; and the undulations are obtained by the repetition of this portion VN of the geodesic line alternately on each side of the equator and of the meridian.

[Figure 1: A diagram of the oblate spheroid showing the vertex V, the meridian VA through V, the equator, and the node N at which the geodesic line meets the equator.]

I consider in the present paper the series of geodesic lines which cut at right angles a given meridian AC, or, say, a series of geodesic normals. It may be remarked that as V passes from the position A on the equator to the pole C, the angular distance AN increases from a certain determinate value (equal, as will appear, to $\frac{C}{A}\cdot 90°$, if C, A are the polar and equatorial axes respectively) up to the value $90°$; and it thus appears that, attending only to their course after they first meet the equator, the geodesic normals have an envelope resembling in its general appearance the evolute of an ellipse (see fig. 1 and also fig. 2), the centre hereof being the point B at the distance $BA = 90°$, and the axes coinciding in direction with the equator BA and meridian BC: this is in fact a real geodesic evolute of the meridian CA. The point α is, it is clear, the intersection of the equator by the geodesic line for which V is consecutive to the point A (so that $\angle BOA = \left(1 - \frac{C}{A}\right) 90°$); and the point γ is the intersection of the meridian CB by the geodesic line for which V is consecutive to the point C; and its position will be in this way presently determined. I was anxious, with a view to the construction of a drawing and a model, to obtain some numerical results in relation to a spheroid of considerable excentricity, and I selected that for which

$$\frac{C}{A} = \tfrac{1}{2} \quad \text{(polar axis} = \tfrac{1}{2} \text{ equatorial).}$$

[Figure 2: A diagram showing the geodesic evolute of the meridian, with the point B, the vertex region, the points α on the equator and γ on the meridian CB.]

Before proceeding further, I remark that Legendre's expression "reduced latitude" is used in what is not, I think, the ordinary sense; and I propose to substitute the term "parametric latitude": viz., in fig. 3, referring the point P on the ellipse by means of the ordinate MPQ to a point Q on the circle, radius $OK\,(= OA$, fig. 1), and drawing the normal PT, then we have for the point P the three latitudes,

$$\begin{aligned} \lambda &= \angle PTK, && \text{normal latitude,} \\ \lambda'' &= \angle POK, && \text{central latitude,} \\ \lambda' &= \angle QOK, && \text{parametric latitude;} \end{aligned}$$

[1] *Mém. de l'Inst.* 1806; see also the *Exer. de Calcul Intégral*, t. I. (1811), p. 178, and the *Traité des Fonctions Elliptiques*, t. I. (1825), p. 360.

viz. λ' is the parameter most convenient for the expression of the values of the coordinates x, y ($x = A\cos\lambda'$, $y = C\sin\lambda'$) of a point P on the ellipse. The relations between the three latitudes are

$$\tan\lambda'' = \frac{C}{A}\tan\lambda' = \frac{C^2}{A^2}\tan\lambda,$$

so that λ'', λ', λ are in the order of increasing magnitude. I use in like manner l, l', l'' in regard to the vertex V. The course of a geodesic line is determined by the equation

$$\cos\lambda'\sin\alpha = \text{const.},$$

where λ' is the reduced latitude of any point P on the geodesic line, and α is at this point the azimuth of the geodesic line, or its inclination to the meridian. Hence, if l' be the parametric latitude of the vertex V, the equation is

$$\cos\lambda'\sin\alpha = \cos l'$$

(whence also, when $\lambda' = 0$, $\alpha = 90^\circ - l'$; that is, the geodesic line cuts the equator at an angle $= 90^\circ - l'$, the parametric latitude of the vertex). The equation in question, $\cos\lambda'\sin\alpha = \cos l'$, leads at once to Legendre's other equations: viz. taking, as above, A, C for the equatorial and polar semiaxes respectively, and δ for the excentricity, $\delta = \sqrt{1 - \dfrac{C^2}{A^2}}$; and to determine the position of P on the meridian, using (instead of the parametric latitude λ') the angle ϕ determined by the equation

$$\cos\phi = \frac{\sin\lambda'}{\sin l'},$$

and writing, moreover, s to denote the geodesic distance VP, and Λ to denote the longitude of P measured from the meridian CA which passes through the vertex V, these are

$$\begin{aligned} ds &= d\phi\sqrt{C^2 + A^2\delta^2\sin^2 l'\cos^2\phi}, \\ d\Lambda &= \frac{\cos l'}{A}\cdot\frac{d\phi\sqrt{C^2 + A^2\delta^2\sin^2 l'\cos^2\phi}}{1 - \sin^2 l'\cos^2\phi}, \end{aligned}$$

which differential expressions are to be integrated from $\phi = 0$; and the equations then determine λ', s, and Λ, all in terms of the angle ϕ,—that is, virtually s and Λ, the length and longitude, in terms of the parametric latitude λ'.

Writing, with Legendre,

$$\begin{aligned} c^2 &= \frac{A^2\delta^2\sin^2 l'}{C^2 + A^2\delta^2\sin^2 l'}, \quad = \delta^2\sin^2 l, \\ b^2 &= 1 - c^2 = \frac{C^2}{C^2 + A^2\delta^2\sin^2 l'}, \quad = 1 - \delta^2\sin^2 l; \end{aligned}$$

also

$$n = \tan^2 l', \qquad M = \frac{C}{bA\cos l'} = \frac{C}{A\cos l},$$

then the formulæ become

$$\begin{aligned} ds &= \frac{C}{b}\,d\phi\sqrt{1 - c^2\sin^2\phi}, \\ d\Lambda &= M\,\frac{d\phi\sqrt{1 - c^2\sin^2\phi}}{1 + n\sin^2\phi}. \end{aligned}$$

Hence integrating from $\phi = 0$, and using the notations F, E, Π of elliptic functions, we have

$$s = \frac{C}{b} E(c, \phi),$$
$$\Lambda = \frac{M}{n}\big\{(n + c^2)\,\Pi\,(n, c, \phi) - c^2 F(c, \phi)\big\};$$

viz. these belong to any point P whatever on the geodesic line, parametric latitude of vertex $= l'$; and if we write herein $\phi = 90°$, then they will refer to the node N, or point of intersection with the equator.

The position of the point α is at once obtained by writing $l' = 0$: viz. this gives $c = 0$, $b = 1$, $M = \dfrac{C}{A}$, $n = 0$: the differential expressions are $ds = C\,d\phi$, $d\Lambda = \dfrac{C}{A}\,d\phi$. Or integrating from $\phi = 0$ to $\phi = \frac{1}{2}\pi$, we have $s = A \cdot \dfrac{C}{A} \cdot \frac{1}{2}\pi$, $\Lambda = \dfrac{C}{A} \cdot \frac{1}{2}\pi$, agreeing with each other, and giving longitude of $\alpha = \dfrac{C}{A} \cdot \frac{1}{2}\pi$; or, what is the same thing, $\angle \alpha OB = \frac{1}{2}\pi\left(1 - \dfrac{C}{A}\right)$.

Writing in the formulæ $l' = 90°$, we have $c = \delta$, $b = \dfrac{C}{A}$, $\dfrac{M}{n} = 0$; whence $d\Lambda = 0$, or $\Lambda = \text{const.} = \frac{1}{2}\pi$, since the geodesic line here coincides with the meridian CB; and moreover $s = A\,E(\delta, \phi)$; viz. this is merely the expression of the distance from C of a point P on the meridian CB. But we do not thus obtain the position of the point γ.

To find it we must consider a position of V consecutive to C, say, $l' = \frac{1}{2}\pi - \varepsilon$, where ε is indefinitely small; n is thus indefinitely large, and the integral $\Pi\,(n, c, \phi)$ is not conveniently dealt with. But it may be replaced by an expression depending on $\Pi\!\left(\dfrac{c^2}{n}, c, \phi\right)$, where $\dfrac{c^2}{n}$ is indefinitely small; viz. (Legendre, *Fonct. Ellipt.* vol. I. p. 69) we have

$$\Pi\,(n, c, \phi) = F(c, \phi) + \frac{1}{\sqrt{a}}\,\tan^{-1}\frac{\sqrt{a}\,\tan\phi}{\sqrt{1 - c^2\sin^2\phi}} - \Pi\!\left(\frac{c^2}{n}, c, \phi\right),$$

where

$$a = (1 + n)\left(1 + \frac{c^2}{n}\right).$$

We thus have

$$\Lambda = \frac{M}{n}\left\{nF(c, \phi) + \frac{\sqrt{n}\,\sqrt{c^2 + n}}{\sqrt{1 + n}}\,\tan^{-1}\frac{\sqrt{a}\,\tan\phi}{\sqrt{1 - c^2\sin^2\phi}} - (c^2 + n)\,\Pi\!\left(\frac{c^2}{n}, c, \phi\right)\right\},$$

where, $\dfrac{c^2}{n}$ being small,

$$\Pi\!\left(\frac{c^2}{n}, c, \phi\right) = \int \frac{d\phi}{\left(1 + \dfrac{c^2}{n}\sin^2\phi\right)\sqrt{1 - c^2\sin^2\phi}}$$
$$= \int \frac{\left(1 - \dfrac{c^2}{n}\sin^2\phi\right)d\phi}{\sqrt{1 - c^2\sin^2\phi}} = \left(1 - \frac{1}{n}\right)F(c, \phi) + \frac{1}{n}\,E(c, \phi).$$

And expanding also the $\tan^{-1}$ term, we thus have

$$\begin{aligned}\Lambda &= \frac{M}{n}\left\{nF(c,\,\phi) + \frac{\sqrt{n}\,\sqrt{c^2+n}}{\sqrt{1+n}}\left[\tfrac{1}{2}\pi - \frac{\sqrt{1-c^2\sin^2\phi}}{\tan\phi}\cdot\frac{\sqrt{n}}{\sqrt{(1+n)(c^2+n)}}\right]\right.\\ &\qquad\left. -\,(c^2+n)\left[\left(1-\frac{1}{n}\right)F(c,\,\phi) + \frac{1}{n}\,E(c,\,\phi)\right]\right\}\\ &= \frac{M}{n}\left\{\left(b^2+\frac{c^2}{n}\right)F(c,\,\phi) - \left(1+\frac{c^2}{n}\right)E(c,\,\phi) + \frac{\sqrt{n}\,\sqrt{c^2+n}}{\sqrt{1+n}}\cdot\tfrac{1}{2}\pi\right.\\ &\qquad\left. -\,\frac{n}{n+1}\cot\phi\,\sqrt{1-c^2\sin^2\phi}\right\},\end{aligned}$$

which, in the term in {} neglecting negative powers of n, becomes

$$\Lambda = \frac{M}{n}\big\{\sqrt{n}\cdot\tfrac{1}{2}\pi + b^2F(c,\,\phi) - E(c,\,\phi) - \cot\phi\,\sqrt{1-c^2\sin^2\phi}\big\}.$$

We may moreover write $c=\delta$, $b=\dfrac{C}{A}$, $\phi = 90^\circ - \lambda'$, $n=\dfrac{1}{\varepsilon^2}$, $M=\varepsilon$, and therefore $\dfrac{M}{n}=\varepsilon$, so that the formula is

$$\begin{aligned}\Lambda &= \varepsilon\left\{\frac{1}{\varepsilon}\cdot\tfrac{1}{2}\pi + b^2F(c,\,90^\circ-\lambda') - E(c,\,90^\circ-\lambda') - \tan\lambda'\,\sqrt{1-c^2\cos^2\lambda'}\right\}\\ &= \tfrac{1}{2}\pi - \varepsilon\big\{\tan\lambda'\,\sqrt{1-c^2\cos^2\lambda'} + E(c,\,90^\circ-\lambda') - b^2F(c,\,90^\circ-\lambda')\big\},\end{aligned}$$

where I retain c, b as standing for $\sqrt{1-\dfrac{C^2}{A^2}}$, $\dfrac{C}{A}$ respectively.

Writing herein $\lambda'=0$, we have

$$\Lambda = \tfrac{1}{2}\pi - \varepsilon\,(E,\,c - b^2F,\,c),$$

where the coefficient $E,\,c-b^2F,\,c$ is

$$= \int_0^{\frac{1}{2}\pi} d\theta\left(\sqrt{1-c^2\sin^2\theta} - \frac{1-c^2}{\sqrt{1-c^2\sin^2\theta}}\right) = c^2\int_0^{\frac{1}{2}\pi}\frac{\cos^2\theta\,d\theta}{\sqrt{1-c^2\sin^2\theta}},$$

consequently positive; that is, Λ, the longitude of the node, is less than 90°, as it should be. Hence in order that Λ may be $= 90^\circ$, we must have λ' negative, say, $\lambda' = -\mu'$, where μ' is positive; and, observing that we may under the signs E, F write $90^\circ-\mu'$ instead of $90^\circ+\mu'$, we thus have

$$\tfrac{1}{2}\pi = \tfrac{1}{2}\pi + \varepsilon\big\{\sqrt{1-c^2\cos^2\mu'}\,\tan\mu' - E(c,\,90^\circ-\mu') + b^2F(c,\,90^\circ-\mu')\big\};$$

that is, we must have

$$\tan\mu'\,\sqrt{1-c^2\cos^2\mu'} = E(c,\,90^\circ-\mu') - b^2F(c,\,90^\circ-\mu');$$

viz. μ' is here the parametric latitude (south) of the intersection of the meridian CB with the consecutive geodesic line—that is, of the point γ. As μ' increases from 0 to 90°, the left-hand side increases from 0 to ∞; and the right-hand side, beginning from a positive value and either attaining a maximum or not, ultimately decreases to 0; there is consequently a real root, which is easily found by trial.

Thus $\dfrac{C}{A}=\tfrac{1}{2}$, $C=\tfrac{1}{2}\sqrt{3}$ (angle of modulus $=60^\circ$), $b=\tfrac{1}{2}$; or the equation is

$$\tan\mu'\,\sqrt{1-\tfrac{3}{4}\cos^2\mu'} = E\,(90^\circ-\mu') - \tfrac{1}{4}F\,(90^\circ-\mu').$$

Using Legendre's Table IX., we have

μ'	$90^\circ-\mu'$	E	F	$E-\frac{1}{4}F$	$\tan\mu'\sqrt{1-\frac{3}{4}\cos^2\mu'}$
0°	90°	1.21105	2.15651	0.6719	0.0
10	80	1.12248	1.81252	0.6693	
20	70	1.02663	1.49441	0.6530	
30	60	0.91839	1.21253	0.6153	0.3819
40	50	0.79538	0.96465	0.5542	0.6278

so that we see the required value is between 30° and 40°; and a rough interpolation gives the value $\mu'=37^\circ\,40'$. But repeating the calculation with the values 37° and 38°, we have

μ'	$90^\circ-\mu'$	E	F	$E-\frac{1}{4}F$	$\tan\mu'\sqrt{1-\frac{3}{4}\cos^2\mu'}$
37°	53°	0.833879	1.035870	0.57419	0.54425
38	52	0.821197	1.011849	0.56823	0.57108

whence, interpolating, $\mu'=37^\circ\,55'$.

The semiaxes of the geodesic evolute, measured according to their longitude and parametric latitude respectively, are thus $B\alpha$, long. of $\alpha=45^\circ$; $B\gamma$, param. lat. $=37^\circ\,55'$. But measuring them according to their geodesic distance, the equatorial radius A being taken $=1$, we have

$$\begin{aligned} B\alpha &= \tfrac{1}{4}\pi = 0.78540,\\ B\gamma &= \left(\frac{C}{b}-1\right)\{E,\, c - E(52^\circ\,5')\} = 1.21106-0.82225 = 0.38881. \end{aligned}$$

Reverting to the general formulæ for s, Λ, but writing therein $A=1$, and therefore $C=\sqrt{1-\delta^2}$; writing also $\phi=90^\circ$ (that is, making the formulæ to refer to the node N of the geodesic line), we have

$$\begin{aligned} s &= \frac{\sqrt{1-\delta^2}}{\sqrt{1-\delta^2\sin^2 l}}\,E,\, c,\\ \Lambda &= \frac{\sqrt{1-\delta^2}}{n\cos l}\big\{(n+c^2)\,\Pi\,(n,\, c) - c^2F,\, c\big\}; \end{aligned}$$

but for the calculation of the second of these formulæ by means of Legendre's Tables it is necessary to express $\Pi\,(n,\, c)$ in terms of the functions E, F.

The proper formula is given in *Fonct. Ellipt.* vol. I. p. 137; viz. this is

$$\frac{\Delta\,(b,\,\theta)}{\sin\theta\cos\theta}\,\Pi\,(n,\, c) = \tfrac{1}{2}\pi + \frac{\sin\theta}{\cos\theta}\,\Delta\,(b,\,\theta)\,F,\, c + F,\, c\,F\,(b,\,\theta) - F,\, c\,E\,(b,\,\theta) - E,\, c\,F\,(b,\,\theta),$$

where $\Delta\,(b,\,\theta)=\sqrt{1-b^2\sin^2\theta}$. θ is an angle given by the equation $\cot\theta=\sqrt{n}$; we have $n=\tan^2 l'$; consequently $\theta=90^\circ-l'$. Substituting this value, except that for shortness I retain $E\,(b,\,\theta)$, $F\,(b,\,\theta)$ in place of $E\,(b,\,90^\circ-l')$, $F\,(b,\,90^\circ-l')$, we have

$$\begin{aligned} \Delta\,(b,\,\theta) &= \sqrt{1-b^2\cos^2 l'}\\ &= \sqrt{1-(1-\delta^2\sin^2 l)\cos^2 l'} = \sin l, \end{aligned}$$

and thence

$$\tan\theta\,\Delta\,(b,\,\theta) = \cot l'\,\sin l = \frac{\cos l}{\sqrt{1-\delta^2}};$$

whence

$$\Pi\,(n,\, c) = \frac{\sin l'\cos l'}{\sin l}\left\{\tfrac{1}{2}\pi + F,\, c\left[\frac{\cos l}{\sqrt{1-\delta^2}} + F\,(b,\,\theta) - E\,(b,\,\theta)\right] - E,\, c\,F\,(b,\,\theta)\right\}.$$

But

$$n + c^2 = \tan^2 l' + \delta^2 \sin^2 l = \sin^2 l \sec^2 l'.$$

Hence

$$(n + c^2)\,\Pi\,(n,\, c) - c^2 F,\, c = \sin^2 l\big\{\sec^2 l'\,\Pi\,(n,\, c) - \delta^2 F,\, c\big\},$$

and multiplying this by $\dfrac{\sqrt{1-\delta^2}}{n\cos l}$, $=\dfrac{\sqrt{1-\delta^2}\cos l}{\tan^2 l'\cos^2 l'}$, the exterior factor is

$$\frac{\sqrt{1-\delta^2}\cos l\tan^2 l'}{\tan^2 l'\cdot\sqrt{1-\delta^2}} = \frac{\cos l}{\sqrt{1-\delta^2}},$$

and we have

$$\Lambda = \frac{\cos l}{\sqrt{1-\delta^2}}\big\{\sec^2 l'\,\Pi\,(n,\, c) - \delta^2 F,\, c\big\},$$

which is the formula I used in the calculations. It would, however, have been better to reduce a step further; viz. we have

$$\begin{aligned}\sec^2 l'\,\Pi\,(n,\, c) &= \frac{\sqrt{1-\delta^2}}{\cos l}\left\{\tfrac12\pi + F,\, c\left[\frac{\cos l}{\sqrt{1-\delta^2}} + F\,(b,\,\theta) - E\,(b,\,\theta)\right] - E,\, c\,F\,(b,\,\theta)\right\}\\ &= \frac{\sqrt{1-\delta^2}}{\cos l}\,\big\{\tfrac12\pi + F,\, c\big[F\,(b,\,\theta) - E\,(b,\,\theta)\big] - E,\, c\,F\,(b,\,\theta)\big\} + F,\, c,\end{aligned}$$

and thence

$$\sec^2 l'\,\Pi\,(n,\, c) - \delta^2 F,\, c = \frac{\sqrt{1-\delta^2}}{\cos l}\big\{\tfrac12\pi + F,\, c\big[F\,(b,\,\theta) - E\,(b,\,\theta)\big] - E,\, c\,F\,(b,\,\theta)\big\} + \sqrt{1-\delta^2}\,F,\, c;$$

or, finally,

$$\Lambda = \tfrac12\pi + F,\, c\,F\,(b,\,\theta) - F,\, c\,E\,(b,\,\theta) - E,\, c\,F\,(b,\,\theta) + \sqrt{1-\delta^2}\cos l\,F,\, c.$$

It is easy with this expression of Λ to obtain the results already found for the extreme values $l' = 0°$, $l' = 90°$.

As Legendre's Tables have for argument, not the modulus c, but the angle of the modulus, say χ (that is, $\sin\chi = c = \delta\sin l$), it is convenient to replace $\sqrt{1-\delta^2\sin^2 l}$ by its value $\cos\chi$; and the formulæ thus are

$$\begin{aligned}s &= \frac{\sqrt{1-\delta^2}}{\cos\chi}\,E,\, c,\\ \Lambda &= \tfrac12\pi + F,\, c\Big[\sqrt{1-\delta^2}\cos l + F\,(b,\,\theta) - E\,(b,\,\theta)\Big] - E,\, c\,F\,(b,\,\theta),\end{aligned}$$

where

$$c = \sin\chi = \delta\sin l, \qquad \tan l' = \sqrt{1-\delta^2}\,\tan l, \qquad \theta = 90° - l';$$

and in the case intended to be numerically discussed, $\delta = \frac12\sqrt3$, $\sqrt{1-\delta^2} = \frac12$. I take l' as the argument, giving it the values $0°$, $10°$, …, $90°$, and perform the calculation as shown in the Table.

[Table: The Table of numerical values of l, χ, $90° - \chi$, $\frac12\cos l$, F, c, E, c, $F\,(b,\,\theta)$, $E\,(b,\,\theta)$, their combinations, Λ in radians and degrees, and the length s, for $l' = 0°$, $10°$, …, $90°$, as displayed on page 23 of the original. Columns marked with an asterisk show respectively the longitude of the node and the length (or distance of node from vertex) for the geodesic lines belonging to the different values of the argument l'.]

The remarks which follow have reference to the stereographic projection of the figure on the plane of the equator, the centre of projection being the pole (say the South Pole) of the spheroid.

It is to be remarked that if a point P of the spheroid is projected as above, by means of an ordinate into the point Q of the sphere radius OK $(= OA)$, then projecting stereographically as to the spheroid and the sphere from the south poles thereof respectively, the points P and Q have the same projection. And it is hence easy to show that an azimuth α at a point of the meridian (parametric latitude λ', normal latitude λ, and therefore $\tan\lambda' = \dfrac{C}{A}\tan\lambda$) is projected into an angle (α) such that

$$\tan(\alpha) = \frac{\sin\lambda'}{\sin\lambda}\tan\alpha.$$

In fact in fig. 3, if we take therein OK, OC for the axes of x, z respectively, and the axis of y at right angles to the plane of the paper, and if we have at P on the surface of the spheroid an element of length PR at the inclination α to the meridian PK, then if x, y, z are the coordinates of P, and $x+\delta x$, $y+\delta y$, $z+\delta z$ those of R, we have

$$\begin{aligned}\delta x &= \rho\cos\alpha\sin\lambda,\\ \delta z &= -\rho\cos\alpha\cos\lambda,\\ \delta y &= \rho\sin\alpha,\end{aligned}$$

and thence

$$\tan\alpha = \frac{\delta y}{\sqrt{\delta x^2+\delta z^2}}.$$

Now, if the meridian and the points P, R are referred by lines parallel to Oz to the surface of the sphere radius OA, the only difference is that the ordinates z are increased in the ratio $C : A$; so that if the projected angle be (α), we have

$$\tan(\alpha) = \frac{\delta y}{\sqrt{\delta x^2+\dfrac{A^2}{C^2}\delta z^2}};$$

and then projecting the sphere stereographically from its south pole, the angle in the projection is $= (\alpha)$. And according to the foregoing remark, the angle (α) thus obtained is also the projection of α from the south pole of the spheroid. We have thus

$$\frac{\tan(\alpha)}{\tan\alpha} = \frac{\sqrt{\delta x^2+\delta z^2}}{\sqrt{\delta x^2+\dfrac{A^2}{C^2}\delta z^2}} = \frac{\sqrt{\sin^2\lambda+\cos^2\lambda}}{\sqrt{\sin^2\lambda+\dfrac{A^2}{C^2}\cos^2\lambda}} = \sqrt{\frac{1+\cot^2\lambda}{1+\cot^2\lambda'}} = \frac{\sin\lambda'}{\sin\lambda},$$

which is the required relation.

The foregoing equations,

$$\cos\lambda'\sin\alpha = \cos l', \qquad \tan\lambda' = \frac{C}{A}\tan\lambda, \qquad \tan(\alpha) = \frac{\sin\lambda'}{\sin\lambda}\tan\alpha,$$

determine in the stereographic projection the inclination (α) to the radius, or projection of the meridian, of the geodesic line (parametric latitude of vertex $= l'$) at the point the parametric latitude of which is $= \lambda'$; viz. they enable the construction (in the projection) of the direction of the successive elements of the geodesic line. There would be no difficulty in performing the construction geometrically; but it would, I think, be more convenient to calculate (α) numerically for a given value of l' and for the successive values of λ'. Observe that for $\lambda' = 0$ we have (as above) $90^\circ - \alpha = l'$, and then $\dfrac{\sin\lambda'}{\sin\lambda} = \dfrac{\tan\lambda'}{\tan\lambda} = \dfrac{C}{A}$, consequently $\tan(\alpha) = \dfrac{C}{A}\cot l'$: but we have also $\cot l' = \dfrac{A}{C}\cot l$, so that this equation becomes $\tan(\alpha) = \cot l$, or we have $90^\circ - (\alpha) = l$; viz. in the projection, the geodesic line cuts the equator at an angle $l =$ the normal latitude of the vertex of the geodesic line.

The preceding formulæ and results have enabled me to construct a drawing, on a large scale, of the stereographic projection of the geodesic lines for the spheroid, polar axis $= \frac{1}{2}$ equatorial axis.

[End of pages 1–25. Paper 422 continues to pages 26 ff. in subsequent files.]